# Raj Samrendra Kumar

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## Summary

Computer Science student with practical experience in Machine Learning, Python, Data Analysis, and Generative AI. Currently working as an AI/ML Intern at Rubixe on real-world predictive modeling and data science projects. Skilled in RAG, multi-agent systems, and end-to-end machine learning workflows.

## Work Experience

**Rubixe — AI/ML Intern** **Present**
*Remote*

- Working on **Machine Learning capstone projects** involving customer transaction prediction, earthquake damage prediction, and flight price prediction using real-world datasets.
- Developing end-to-end **Data Science workflows** covering preprocessing, feature engineering, model training, evaluation, and model comparison using **Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-Learn**.
- Building and evaluating **classification and regression models**, comparing appropriate performance metrics, and identifying suitable models for potential production deployment.
- Applying Machine Learning techniques to **banking, earthquake damage, and flight pricing** problems, focusing on data quality, feature engineering, model selection, and predictive performance.

## Education

**Parul Institute Of Engineering and Technology** **2023 – 2027**
*B.Tech in Computer Science and Engineering* *CGPA: 7.77/10*

- **Relevant Coursework:** Object-Oriented Programming, DBMS, Operating Systems, Data Structures

## Projects

**Zomato Restaurant Analysis Dashboard** | Power BI, Power Query, Data Analysis | GitHub **2025**

- Analyzed **51K+ Bangalore Zomato restaurant records** to identify patterns in ratings, votes, cuisines, locations, pricing, restaurant types, and service offerings.
- Built an interactive **Power BI dashboard** using Power Query for data cleaning and transformation, with KPIs, slicers, and visualizations for restaurant performance, pricing, cuisine, and location trends.

**J.A.R.V.I.S. — Voice-Controlled Agentic AI OS** | Python, React, FastAPI, WebSockets | GitHub **2026**

- Developed a multi-agent system using a **Groq LLaMA 3.3 70B router** to classify voice/text queries and orchestrate **7 specialized agents** for web, system, live data, website building, and automation.
- Engineered biometric verification using **OpenCV LBPH face recognition** and MFCC-based voice passphrase matching with automated OS locking and intruder event capture.
- Built a real-time **React and WebSocket dashboard** with audio visualization, biometric metrics, and intruder event logs.

**Flowers Knowledge RAG Chatbot** | Python, LangChain, Groq, Hugging Face, ChromaDB, Streamlit | Live **2025**

- Built an end-to-end **RAG pipeline** using LangChain, recursive text splitting, **Hugging Face all-MiniLM-L6-v2 embeddings**, and ChromaDB for semantic document retrieval.
- Developed a Streamlit Q&A application using **Groq LLaMA 3.3 70B**, grounded prompting, and retrieved context to generate knowledge-based responses.

## Technical Skills

**Programming:** Python, Java, C, SQL

**AI / ML:** TensorFlow, Scikit-Learn, NLP, Computer Vision, Generative AI, Agentic AI, AI Agents, Langflow

**Data Science:** Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Model Evaluation

**Frameworks & APIs:** FastAPI, LangChain, LangGraph, Streamlit

**Databases:** MongoDB, ChromaDB

**Tools & Platforms:** Git, Tableau, Power BI, Microsoft Azure

## Certifications

**Microsoft Certified: Azure AI Engineer Associate (AI-102)** (View Certificate) **2025**
**NVIDIA — Building RAG Agents with LLMs** (View Certificate) **2026**
**Oracle Cloud Infrastructure Generative AI Professional** (View Certificate) **2025**